QUANT

Percentages and Profit & Loss

Assignment 01

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Percentage, P & L Online Assignment 01

1. Soni is 20 % heavier than Rekha in weight. By how much percent is Rekha lighter than Soni?

1. 20 % 2. 25 % 3. 16.67 % 4. None of these

2. What is the value of a property, which has been insured at the rate of 1/6% of its value, if the annual premium paid on it is Rs. 83.33 (in Rs.)?

1. 25,000 2. 50,000 3. 75,000 4. 45,000

3. In a class of 72 students, 87.5 % of the students cleared the CAT exam. How many did not clear it?

1. 12 2. 9 3. 18 4. None of these

4. There are 135 students in a school who get scholarship for their education. If they account for 9 % of the students in the school, how many students did not receive scholarship?

1. 1365 2. 1500 3. 1635 4. 1350

5. When the cost of cigarettes increases by 25 %, a man reduces his annual consumption by 10 %. Find the % change in his annual expenditure on cigarettes?

1. 12.5 % 2. 11.11 % 3. 15 % 4. 6.25 %

6. 37 % of Sita's earnings are swallowed up in taxes, and 75 % of the remainder on living expenses. If she has Rs.7,560 left, what was her salary (in Rs.)?

1. 36,000 2. 24,000 3. 11,500 4. 48,000

7. For what sum should an office with furnishings worth Rs. 665,000 be insured at 5 % so that in the event of a loss, the owner recovers not just the value of the office and furnishings but also the premium paid (in Rs.)?

1. 670,000 2. 750,000 3. 700,000 4. 811,000

8. Khan the barber shaved 40 % of his customers and gave a haircut to 80 % of his customers. He charged Rs. 7 for a shave and Rs. 5 for a haircut. If 20 % of customers who opted for a shave also had a hair-cut, what were Khan's earnings if he had 75 customers (in Rs.)?

1. 410 2. 1,020 3. 510 4. None of these

9. 15% of Malati's sales comprised strawberry cones, 20 % mango cones, 12% candies and 159 vanilla cups. How many candies did she sell?

1. 300 2. 141 3. 36 4. 60

10. Roy drove at a speed 30 % greater than usual outside the city limits. If he normally drove at 60 kmph in the city, then how long did he take to do a 624 km stretch on the highway?

1. 20.8 hrs 2. 8 hrs 3. 10.4 hrs 4. 7.3 hrs

Answer key & Explanations

Percentage, P & L Assignment 01		
1.	3	Required % = $\frac{20}{120} \times 100 = 16.67$ %. (Apply $\frac{R}{100+R} \times 100$).
2.	2	Let the value of the property be x. $\therefore x \times \frac{1}{6} \times \frac{1}{100} = 83\frac{1}{3} \Rightarrow x = \frac{250}{3} \times 6 \times 100 \Rightarrow x = 50000$
3.	2	Numbers of students who did not clear the CAT exam = $\frac{12.5}{100} \times 72 = 9$.
4.	1	135 are 9 % of the total students $\therefore$ Total students = $\frac{135}{9} \times 100 = 1500$ $\therefore$ Number of students who did not receive scholarship = 1500 – 135 = 1365.
5.	1	% change in the annual expenditure = 1.25 × 0.9 = 1.125 $\Rightarrow$ Net increase of 12.5 %
6.	4	Let her salary be x. Net amount left = 7560 $\therefore x = 7560 \times \frac{100}{63} \times \frac{100}{25} = 48000$.
7.	3	Let the total sum = x $\therefore 665000 + \frac{5}{100}x = x \Rightarrow x = 700000$.
8.	3	Total customers = 75. Numbers of customers shaved = $75 \times \frac{40}{100} = 30$. Number of customers who got hair cut = $\frac{80}{100} \times 75 = 60$. $\therefore$His total income= (30 × 7) + (60 × 5) = 210 + 300 = 510.
9.	3	15% comprised strawberry 20 % mango candies 12 % candies $\Rightarrow$ Total 47 %. $\therefore$53 % of total sales = 159 vanilla cups. Hence $\frac{53}{100}$(Total sales) = 159. Total sales = 300. $\therefore$Number of candies = $\frac{12}{100} \times 300 = 36$
10.	2	His speed on the highway = 60 × 1.3 = 78 km/hr $\therefore$ Time taken to cover 624 km = $\frac{624}{78} = 8$ hrs.